



Module 10

Stretching and Myofascial Release

The primary reason for stretching is to increase range of motion at the joints. The purported benefits of stretching – or lack thereof – have recently become hot topics for debate within the realms of fitness and sports performance. The focus of these debates includes:

- Does prestretching (stretching prior to exercise) increase, decrease, or have no effect on athletic performance?
- Does stretching prevent injury?
- Do individuals need to warm up prior to stretching?
- How long should a stretch be held and how often should stretching be done?

The generally accepted principle behind stretching is that when a muscle is stretched, muscle neuron activity decreases, thereby relaxing the muscle, which allows for greater ROM because of reduced resistance. For this reason, it is theorized that prestretching a muscle decreases the force production of a muscle while running.

In contrast, a 2013 study found that utilizing a foam roller prior to exercise can increase range of motion without a decrease in force production (516).

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Physiology of stretching
- Types of stretches
- Area of caution
- Primary stretches
- Myofascial release
- How to maintain properly functioning fascia
- Use of foam roller

❖ PHYSIOLOGY

To understand how the principle of stretching works, it is important to define two proprioceptive parts of a muscle, the *Golgi tendon organ* and the *muscle spindle*.

Golgi Tendon Organ (GTO)

The GTO is a nerve receptor located at the junction of a muscle and tendon. During a stretch, the GTO senses tension in the muscle/tendon and sends a signal to the spinal cord. The spinal cord then sends a signal back to the muscle to relax. This is called the *Golgi tendon reflex*. The GTO exists to regulate muscle tension so that injury does not occur to a muscle and its associated connective tissue (365).

Muscle Spindle

Muscle spindles are located deep within a muscle. Muscle spindles sense change in muscle length as well as the range of change in length and send a signal to the spine. This signal triggers the stretch reflex (also called *myotatic reflex*). The stretch reflex resists the increase in muscle length by causing a muscle to contract (365). The more rapidly muscle length increases, the

stronger the stretch reflex is. Therefore, the primary function of muscle spindles is to protect the body from injury due to overstretching.

The goal of stretching is to minimize the stretch reflex so that a muscle can be lengthened.

ELASTIC RESPONSE

As noted in the *Running Mechanics* module, the stiffer the *muscle-tendon unit* is, the greater the elastic response. In other words, the greater the stiffness of the *muscle-tendon unit*, the greater the potential energy source is. However, if an individual's flexibility and range of motion do not match the degree of flexibility required for a particular movement, performance will decrease and the potential for injury will increase. Therefore the amount of flexibility and range of motion in a particular area should match the amount of flexibility required for the activity being participated in. Substantially more or less flexibility than is required will negatively impact an individual. In particular, too much range of motion often equates to a lack of neuromuscular control for a given area as well as a decrease in force production (608).

Much of the research pertaining to stretching and performance enhancement has been based on different parameters, and thus the results of some of the studies contradict the results of others. For example, some studies show that prestretching a muscle prior to exercise does not hamper performance, while other studies show that it does as a result of decreased force production (111, 241, 242). It should be noted that many of the studies citing a decrease in force production are examining maximal force production, which is not typical for distance runners. Therefore, it could be theorized that prestretching is not as much of an issue for endurance sports as it would be for sports that predominantly require short, maximal efforts (ex: sprinting).

❖ TYPES OF STRETCHES

There are four primary types of stretches:

1. Static

An individual performs and holds a stretch at the end range of motion (ROM).

- **Active:** individuals perform the stretch themselves
- **Passive:** an external force (i.e., another person) acts upon an individual's body part being stretched

When holding a stretch for a relatively long period (> 40 seconds), the muscle spindles become accustomed to the new increase in length and reduce the signals sent to the spinal cord to tell the muscles to contract.

2. Dynamic

An individual moves a body part at a controlled speed through a ROM that mimics the motions performed while running.

3. Ballistic

Similar to the *dynamic stretch*, but the movement is done with momentum with the goal to stretch past the current ROM limit. This is often considered a dangerous stretch and is not recommended for use with clients. Because of the speed with which a proper ballistic stretch is executed, the stretch reflex is triggered and therefore greatly limits the ability of a muscle to stretch.

4. Proprioceptive Neuromuscular Facilitation (PNF)

A PNF stretch consists of a passive stretch followed by an isometric muscle contraction. A partner passively moves a body part to its ROM limit and then the client isometrically contracts the muscle being stretched for 10–15 seconds before relaxing the muscle briefly. The partner then passively moves the body part to a new ROM. This cycle is repeated two or three times in total.

PNF stretching works because of the effect it has on the *Golgi tendon organ*. During PNF stretching, a muscle is repeatedly contracted and relaxed, which triggers the Golgi tendon reflex numerous times. This expands the ROM of the targeted body part beyond what non-PNF stretches can accomplish (114).

AREA OF CAUTION

Overstretching a muscle and/or connective tissue can cause injury. No matter how skilled you are or how long you have been stretching clients, it is impossible to be certain of the integrity of a muscle and its connective tissue. Many coaches and fitness professionals “go by feel” when stretching, which is one of the contributing factors to the injury rate associated with assisted stretching. Just because you feel substantial resistance during a stretch does not necessarily mean it is the end ROM, or conversely, you could have passed the end ROM and caused an injury. It is very important when performing passive stretching or PNF stretching that throughout the stretch you consistently solicit verbal feedback about the muscle tension your client is feeling.

Let your clients dictate what the end ROM is, not you. By pushing past what your clients communicate is the end ROM or by trying to guess the end ROM yourself, you risk injuring your clients. Before you start the stretch, let your clients know what stretch you will be doing and why. Also, tell them that they will need to tell you when they feel the muscle has been stretched to the end ROM. Whether or not at the actual end ROM, you must respect your clients’ feedback and stop when they tell you to.

If a client has an injury or a suspected injury such as a strain, the individual should not stretch or perform myofascial release on the injured muscle, as this may make the injury worse. With any injury or suspected injury, the client’s first course of action should always be to seek out the advice of a specialist.

The safest type of stretch for your clients to perform is static, active stretching (self-stretch). This is beneficial because they can do this on their own. While clients could injure themselves stretching during a static active stretch, the likelihood is minimal.

If your client is experiencing muscle pain that is greater than normal or the result of *delayed onset muscle soreness* (DOMS), do not perform or prescribe any stretches or myofascial release. Refer your client to a specialist such as a physical therapist. Additionally, if a client has a postural issue that you feel could impact performance or safety, it is strongly advised to refer the person to a specialist.

❖ PRIMARY STRETCHES

The stretches that follow are all self-stretches. Instruct your client to hold each stretch for a minimum of 30–40 seconds.

UPPER BODY

Pectoralis Major

- Keep a slight bend in the elbow.



Figure 10.1 Pectoralis major stretch

Latissimus Dorsi

- Make sure hands are on top of each other.



Figure 10.2 Latissimus dorsi stretch

Rhomboids/Lower Trapezius



Figure 10.3 Rhomboids/lower trapezius stretch

- Allow the shoulder blades to rotate forward (the upper back will flex).
- Do not overstretch, but rather hold the stretch as soon as it is felt.

Tricep



- Gently pull the elbow toward midline of the body.

Figure 10.4 Tricep stretch

Levator Scapulae

- Tilt head to side and slightly forward.



Figure 10.5 Levator scapulae stretch

Biceps

- Keep palms facing upward.
- Keep torso as erect as possible while elevating arms behind the body.



Figure 10.6 Biceps stretch

Rotator Cuff

Supraspinatus

Pull arm behind body. Keep scapula depressed.



Figure 10.7 Supraspinatus stretch

Infraspinatus/Teres Minor

Slowly rotate the arms forward so the elbows are pointed as forward as possible in the sagittal plane.



Figure 10.8 Infraspinatus/Teres Minor stretch

Subscapularis

Slowly lower the forearm (as if you're losing at arm wrestling) laterally in the frontal plane. With the elbow on a stable surface, you can also use the opposing arm to facilitate the stretch.

Lower the arm until slight tension is felt. Repeat with the other arm.



Figure 10.9 Subscapularis stretch

LOWER BODY



Figure 10.10 Hamstring stretch

Hamstrings

This can be modified by propping one leg up on an elevated object such as a chair and bending forward at the hips. Additionally, this stretch can be done seated.

Gluteus Maximus

Pull the leg toward the midline of the body.



Figure 10.11 Gluteus maximus stretch



Piriformis/Gluteus Medius

Keep the tibia of the leg crossed as close to horizontal as possible.

Bend forward at the hips.

Figure 10.12 Piriformis/Gluteus medius stretch

Gastrocnemius

Stretch slowly to avoid overstretching the Achilles tendon.



Figure 10.13 Gastrocnemius stretch



Soleus

Maintain 90-degree knee bend during stretch.

Stretch slowly to avoid overstretching the Achilles tendon.

Figure 10.14 Soleus stretch

Hip Adductors

Keep pelvis horizontal during stretch.



Figure 10.15 Hip adductor stretch

Hip Flexors (Iliopsoas)

Keep the upper body as vertical as possible with the arms overhead.



Figure 10.16 Hip flexor stretch

By performing this stretch with the arms overhead, the *pectoralis major* and *latissimus dorsi* are also stretched.



Obliques/Quadratus Lumborum

Keep knees locked out. Shift hips to the opposite side of the upper-body lean.

Figure 10.17 Obliques/Quadratus Lumborum stretch

Rectus Abdominis

This stretch places the lumbar spine in extension.

DO NOT perform this stretch if lumbar extension is contraindicated.



Figure 10.18 Rectus abdominis stretch



Thoracic Spine Musculature

Rotate through the thoracic spine, not the lumbar spine.

Start



End

Figure 10.19 Thoracic spine musculature stretch

❖ MYOFASCIAL RELEASE

Myofascial release (MFR) is a term often used within the sports and fitness worlds. It is commonly used to describe a process in which tight muscles have compression applied to them to “release” tightness, thus restoring proper function (115). To understand exactly what MFR is, let us examine the name. *Myo* is a prefix that means *muscle* (333), and *fascial* relates directly to the *fascia* of the body, as described in an earlier module. Therefore, the term *myofascial release* pertains to the reduction of tightness and restrictiveness of muscles and fascia.

Fascia and the role it plays in supporting and mobilizing the body has become known only in the last five or six years (330). Consequently, many coaches and health professionals do not fully understand the role that fascia plays in the body. Therefore, when MFR is noted and/or performed, it is usually done from the standpoint of reducing muscle tightness while overlooking the fascia element.

Fascia is a web-like connective tissue that surrounds all organs, muscles, nerves, blood vessels, and bones in the body. Muscles do not function in isolation of the fascia (330). If a muscle is too tight, the fascia surrounding that muscle and likely fascia in other areas will also be restricted. It is important to reiterate that fascia can also function independently of the muscles (270).

When athletes use the term *myofascial release*, they typically refer to it in the context of either massage (self or performed by a specialist) or utilizing a tool such as a *foam roller* or *massage stick*.

MAINTAIN PROPERLY FUNCTIONING FASCIA

The following information by Julia Lucas provides valuable insight on how to properly care for the body and, more specifically, the fascia (330).

- **Let the body heal if coming back from injury – don’t rush back into training:** If one’s body mechanics are altered due to an injury, the fascia will adapt to new movement (incorrect) patterns.
- **Stretch gently:** Hold stretches for three to five minutes and do not force a stretch.

- **Stay hydrated:** Proper hydration levels are necessary for the fascia to move unrestricted.
- **Move around:** Adhesions can form between fascial surfaces that are stationary for long periods. These adhesions can alter correct body mechanics.
- **Foam Roller:** Described below.

Fascia Specialists

If you or your clients feel that they are not seeing the desired results with their stretching/MFR program, they should seek out a fascia specialist. Specialists use various methods (i.e., stretching, hands-on manual bodywork) to work on fascia. Professionals such as massage therapists, chiropractors, physical therapists, and osteopaths are often trained in myofascial therapy.

FOAM ROLLER

The most common and popular tool used to facilitate MFR is the foam roller. A foam roller is essentially a semihard, round tube of *Styrofoam*™. Foam rollers come in varying degrees of hardness as well as lengths. The concept behind a foam roller is that it “rolls” over a tight muscle and/or connective tissue with the purpose of reducing muscle/connective tissue tightness. While foam rollers can be very helpful, they can make a tight muscle or fascia tighter.

The widely accepted theory surrounding foam rollers is that the more it hurts, the better it is working. This could not be further from the truth. The reality is that too much compression overstimulates the muscle spindles (i.e., nervous system) and causes the muscles to shorten in response as a protective measure against injury (274). Additionally, too much compression can damage muscles and connective tissue on a cellular level (*fibroblasts*) and actually increase inflammation rather than reduce it. Individuals typically apply much more pressure than is necessary and, frequently, in the wrong locations (274). Just because pressure is applied to a tender area via a foam roller or by other means, the pain and tightness will not necessarily go away. In some cases, and as noted above, it can actually make it worse, as excessive pressure from the roller can create trauma.

When fascia is irritated it becomes inflamed. Inflammation leads to the fascia shortening and becoming extremely tight in a localized area (115). Reducing fascial tightness requires much lighter pressure in a specific area and for longer periods than is typically utilized with standard foam-rolling protocol.

It is suggested that if your clients utilize a foam roller or handheld massage stick, they apply light to medium amounts of pressure. There should be no pain as a result of too much pressure being applied. Therefore, in relation to working with fascia, the phrase *pain, no gain* is applicable.

Due to the size of foam rollers, they often do not compress just the desired muscle/fascia, but ancillary muscles and connective tissue as well. When dealing with restricted fascia, it is advised to use a well-worn roller that is softer than a new one.

As noted previously, if you believe your client's discomfort is the result of a sprain or strain, it is advised not to perform stretching or use a foam roller. In this situation, it is advised to seek the attention of a specialist such as a physical therapist.

❖ FOAM ROLLER APPLICATIONS

LOWER BODY

Hamstrings



Figure 10.20 Foam Roller: Hamstrings

Glutes



Figure 10.21 Foam Roller: Glutes

Quadriceps



Figure 10.22 Foam Roller: Quadriceps

Tibialis Anterior



Figure 10.23 Foam Roller: Tibialis anterior

Hip Adductors



Figure 10.24 Foam Roller: Hip adductors

Iliotibial Band (IT Band)



Figure 10.25 Foam Roller: Iliotibial band

Calves (Gastrocnemius/Soleus)



Figure 10.26 Foam Roller: Calves

UPPER BODY

Latissimus Dorsi



Figure 10.27 Foam Roller: Latissimus dorsi

Low Back (Erector Spinae/Quadratus Lumborum)



Figure 10.28 Foam Roller: Low back

Upper Back (Rhomboids/Lower-Middle Trapezius)



Figure 10.29 Foam Roller: Upper back

- Tilt the body slightly to the side so that the roller is not directly on the spine but rather on the back muscles.

❖ SUMMARY

- The *Golgi tendon organ* (GTO) and *muscle spindles* work together to influence the amount of stretch placed on a muscle.
- There are four types of stretches:
 - Static
 - Dynamic
 - Ballistic
 - Proprioceptive neuromuscular facilitation (PNF)
- Be careful not to overstretch a muscle.
- The safest way for clients to stretch is for them to stretch themselves.
- When stretching statically, a stretch should be held for at least 40 seconds to allow the muscle spindles to reduce their signaling, thus relaxing the muscle and allowing for greater range of motion.
- A muscle-tendon unit that has too much flexibility may result in decreased force production.
- *Myofascial release* is a term that typically refers to compression of muscles for the purposes of reducing tightness.
- When fascia is irritated, it becomes inflamed and thus shortens, causing muscle tightness and pain.
- Fascia can be damaged when too much pressure is applied to it.
- Foam rolling prior to exercise has been shown to increase range of motion without a decrease in force production.